THEORY OF COMPUTATION (CS302)

Chomsky Normal Form (CNF)

Learning Objectives

Understand Chomsky Normal Form, its rules, conversion procedure, and applications in context-free grammars.

Introduction to CNF

Chomsky Normal Form (CNF) is a standardized form of a Context-Free Grammar in which every production follows specific rules. CNF simplifies grammar processing and parsing algorithms.

Need for Chomsky Normal Form

• Simplifies grammar analysis
• Helps in parser construction
• Useful for proving theoretical results
• Required by certain parsing algorithms

Rules of CNF

Every production must be of the form:

$A \rightarrow BC$
$A \rightarrow a$

Where:
A, B, C are non-terminals
a is a terminal symbol

Steps for Conversion to CNF

Step 1: Remove null productions.
Step 2: Remove unit productions.
Step 3: Remove useless symbols.
Step 4: Convert remaining productions into CNF format.

Removing Null Productions

Null productions are productions that generate ε. These productions must be eliminated while preserving the language generated by the grammar.

Removing Unit Productions

Unit productions are productions of the form $A \rightarrow B$. They are removed by replacing them with equivalent non-unit productions.

Removing Useless Symbols

Symbols that do not contribute to generating terminal strings or are unreachable from the start symbol are removed.

Worked Example

Original Grammar:
S → AB | a
A → a
B → b

This grammar already satisfies CNF because:
S → AB (A → BC form)
A → a (terminal production)
B → b (terminal production)

Advantages of CNF

• Standardized grammar representation
• Easier parsing
• Useful in theoretical proofs
• Supports CYK parsing algorithm

Applications

• Compiler Design
• Syntax Analysis
• Parsing Algorithms
• Formal Language Theory

Exam Important Questions

1. Define Chomsky Normal Form.
2. State the rules of CNF.
3. Explain the conversion procedure to CNF.
4. What are null productions?
5. Explain unit production removal.
6. Convert a grammar into CNF.

Summary

Chomsky Normal Form provides a standard representation for context-free grammars. It simplifies grammar processing and is widely used in compiler design and formal language theory.